

EECS 570

Midterm Review

Parallel Computer Architecture

Fall 2025

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<http://www.eecs.umich.edu/courses/eecs570/>

Slides developed in part by Profs. Austin, Adve, Falsafi, Martin, Narayanasamy, Nowatzky, Peh, and Wenisch of CMU, EPFL, MIT, UPenn, U-M, UIUC



Intel Paragon XP/S

Midterm logistics

Syllabus: PA1, all lectures and reading until lecture on 10/6

You can bring notes written on 1 A4 sheet (front and back)

Time: 6-7:30p, Wed, Oct 15th

Location:

First letters of uniquname	Room
aa - dr	COOL 1940
gu - io	COOL 2918
ja - zo	COOL G906

SSDs: Same starting time and location

Alternate exam:

Time: 8-9:30am, Thursday, Oct 16th

Location: 4901 BBB

OH: I will hold office hours during fall break on 11am Tuesday, Oct 10/5

Announcements

- No reading/quiz for this week's lectures
- PA1 grades released

1: Intro

- ILP Wall
- Power Wall
- Dennard Scaling / Post-Dennard Scaling
- Motivating Multiprocessors

2: Models and MPI

- Complexity analysis
 - ▣ Work & Depth
 - ▣ Critical Path
- Scheduling
 - ▣ Greedy scheduler
- Amdahl's Law
- Message Passing
 - ▣ Asynchronous vs Synchronous
 - ▣ Deadlock

3: Shared Memory + DLP

- Global address space
- Virtual addressing
- Synchronization
 - Locks
 - Barriers
- Bus based interconnect
- Point-to-point interconnects
- Vector Processing

4: Synchronization

- Acquire/Wait/Obtain
- Locks
 - ❑ Need atomic operations for Read-Modify-Write
 - Test&set
 - Fetch&op
 - Swap
 - Compare&swap
 - ❑ Test and Set Spin
 - ❑ Test and Test and Set Spin
 - ❑ Ticket Locks
 - ❑ Array Based Queue Locks
 - ❑ MCS Lock
- Barriers
 - ❑ Sense reversing barrier
 - ❑ Tree-Based

5: Transactional Memory

- Insert, lookup, delete -> transfer
 - Fine grained locking – difficult to program
 - Coarse grained locking – too slow
 - Create TM (Read set, Write set)
 - Version management (Eager or Lazy)
 - Conflict Detection (Eager or Lazy)
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- Where does TM not work at replacing locks

6: Snooping

- Single Reader Multiple Writer – or – Data value invariant
- Caches snoop all requests on single bus
- Make decisions about cacheline state
- Valid – Invalid
- M – O – E – S – I
- Update vs. Invalidate protocols

7: SMP Designs

- Coherence Control Implementation
- Writebacks, non-atomicity, serialization/order
- Hierarchical caches, inclusive/exclusive caches
- Split Busses
- Deadlock, livelock & starvation
- TLB Coherence

8: Directory Protocols

- Snooping has bus bandwidth and snooping bandwidth problems
- Make a NoC, and add directory protocol
- Centralized vs Distributed
- Share list options
 - Bit vector, pointers, linked-lists, coarse vectors
- 4-hop vs 3-hop transactions
- Use of Ack's
- Race conditions
- Bounding messages

9: Memory Consistency Models

- Type of consistency models
 - ❑ Sequential consistency
 - ❑ TSO
 - ❑ Relaxed consistency
- Language-level vs hardware-level consistency models
- Data-race-free memory model
- Speculative hardware optimizations for memory-ordering overhead
 - ❑ In-window speculation
 - ❑ Out-of-window speculation
 - ❑ Type-based (local vs shared) optimization
- SC-preserving compiler

Go through problems from sample midterms posted on the website